PILOT MILL

PILOT MILL FOR TUBULARS AND LINERS

MILLING

GENERAL DESCRIPTION

MILLING TOOL WITH A GUIDE NOSE (PILOT) THAT ENTERS THE TUBULAR BEING MILLED AND KEEPS THE BLADES CENTRED ON IT. IT IS USED TO MILL WASHPIPE, SAFETY JOINTS, CROSSOVER SWAGES, WASHOVER SHOES, ADAPTERS, CASING, LINERS, DRILL PIPE AND SWAGED CASING. LINER HANGERS CAN BE MILLED EFFICIENTLY, AVOIDING INSIDE CUTS AND SPEAR RUNS. THE NOSE CAN BE DRESSED TO MILL ANY JUNK MET ALONG THE WAY.

TECHNICAL CHARACTERISTICS

Blade O.D.	Top pin conn. (API reg.)	Pilot O.D.	Overall length	Fishing neck length	Wt (lb)
3-1/4 - 3-7/8	2-3/8	1-3/4 - 2-3/4	27	12	40
4-1/8 - 4-3/8	2-3/8	1-3/4 - 2-3/4	27	12	45
4-1/8 - 5-3/8	2-7/8	2-1/8 - 3-1/4	27	12	120
5-1/2 - 5-5/8	3-1/2	2-1/2 - 4-3/4	38	16	240
5-3/4 - 7-3/8	3-1/2	2-1/2 - 4-3/4	38	16	255
6-1/8 - 9-7/8	4-1/2	4-3/4 - 6-3/4	42	18	305

Dimensions in inches, weights in lb (approximate). Standard API regular pin. Other sizes on request.

DESIGN FEATURES

- **Guide nose:** the pilot sits in the tubular bore and centres the blades.
- **Blade size:** about 1/4 in. over the O.D. of the tool joint or coupling to be milled.
- **Pilot size:** same as the drift diameter of the tubular.
- **Dressable nose:** can be dressed to mill junk.
- **Fishing neck:** standard neck for handling and recovery.
- **Typical jobs:** washpipe (most efficient tool for stuck washpipe), liners and hangers, adapters, swaged casing, drill collars with open I.D.

HOW TO ORDER

Please specify: 1. size and weight of the casing to be milled 2. size and weight of the casing to run through, if known 3. top pin connection.

OPERATING PARAMETERS

Rotary speed (start)	80 - 100 rpm
Tool weight (start)	2,000 - 6,000 lb or less
H-40 / K-55 casing	4,000 - 6,000 lb; 80 - 100 rpm
N-80 / P-110 / Q-135 casing	8,000 - 10,000 lb; 100 - 120 rpm
Entering the tubular	about 30 rpm, 2,000 lb
Typical milling rate (casing)	8 - 12 ft/hr

OPERATING NOTES

- **Alignment check:** rotate at about 125 rpm above the tubular to find neutral torque, then enter the pilot slowly; a gradual slow-down when rotation stops shows good alignment.
- **Steady feed:** apply weight at a constant rate, do not drill off.
- **Cuttings:** below a shoe, ream up and down every 15 - 20 ft milled; also in holes deviated 30 degrees or more.
- **Rate drop:** may be a loose steel ring turning with the mill; spud gently. If cutting stops with no torque rise, a tubular section may be turning: pull and retrieve with a spear.
- **Wear pattern:** hook wear means efficient cutting; taper wear means ease off the weight.